Utkarsh Raj

Bachelor of Technology
Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., CSE	National Institute of Technology, Patna	8.49	June,2026
Senior Secondary	The Gurukul Nilokheri/CBSE	95.8	May,2023
Secondary	The Gurukul Nilokheri/CBSE	94.8	August,2021

Projects

- **Project A: MoneyManager – A Spring Boot-based Personal Finance Management System** *September 2026*
  Tools: Java, Spring Boot, Spring Security, JPA, Hibernate, MySQL, JWT, Maven, Git, GitHub [GitHub]
  - Developed a personal finance management application with RESTful APIs for managing income, expenses, categories, user profiles, and transactions.
  - Implemented secure authentication and authorization using Spring Security and JWT, with password encryption and protected API endpoints.
  - Designed the backend using Spring Boot, Spring Data JPA, Hibernate, and MySQL for efficient database operations and persistent financial data management.
  - Integrated Brevo SMTP for email-based account activation and implemented dashboard and filtering APIs for financial summaries and transaction tracking.

- **Project B: Student Management System – Full-Stack Web Application** *August 2026*
  Tools: React, Java, Spring Boot, Spring Data JPA, MySQL, Hibernate, Maven, IntelliJ IDEA, VS Code, Git [GitHub]
  - Developed a full-stack Student Management System using React for the frontend and Spring Boot for the backend, with MySQL for persistent data storage.
  - Designed and implemented RESTful APIs using Spring Boot and Spring Data JPA to perform CRUD operations for student records.
  - Built a responsive React-based user interface for adding, viewing, updating, and deleting student information.
  - Integrated the React frontend with the Spring Boot backend using Axios for seamless client-server communication.
  - Utilized Hibernate and JPA for object-relational mapping and efficient interaction with the MySQL database.

- **Project C: Online Shopping Cart Simulator – Object-Oriented Java Application** *November 2025*
  Tools: Java, OOP Concepts, Collections Framework, File Handling, Exception Handling, VS Code, Git [GitHub]
  - Designed and developed a console-based shopping cart system using core Object-Oriented Programming principles such as encapsulation, inheritance, and polymorphism.
  - Implemented modular classes for Product, User, Cart, and Order management to ensure clean architecture and code reusability.
  - Utilized Java Collections Framework (ArrayList, HashMap) for efficient storage, retrieval, and manipulation of product and cart data.
  - Integrated file handling for persistent storage of user and order details, along with exception handling to ensure robust execution.
  - Simulated real-world e-commerce operations including product browsing, cart updates, checkout process, and bill generation.

Skills

- **Programming Languages**: C,C++,Python,Java,JavaSript,HTML,CSS
- **Technologies**: Spring Boot,React.js,MySql,MongoDB,PostgreSql
- **Tools**: VS Code,Intellij Idea,Git,GitHub,Jupyter Notebook,Flex,Bison,MS Word,Powerpoint,TinkerCad

Certifications

- **NIT Patna ByteVerse 7.0 Hackathon**, Certification 1 *April 2025*

Achievements

- Solve 250+ problems on Leetcode